

First Response & Evidence Handling

Lab Project 01 — Day 03

Name: Mohammad Salem Hassani

Student ID: 2545880

Course: Incident Management

Instructor: Asen Grozdanow

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1. Executive Summary

On 28 April 2026 at 13:51 UTC, a suspected ransomware incident was reported at NorthBridge Health Cooperative affecting the Ubuntu workstation of office manager Anna. The on-call responder immediately secured the digital crime scene by isolating the Target VM from the public internet using an internal-only network named crime-scene. Volatile and non-volatile evidence was collected in strict RFC 3227 order of volatility: six volatile artefacts including a full 2 GB RAM image acquired via LiME (Linux Memory Extractor), and six non-volatile artefacts including system logs, file metadata, and three encrypted file samples with anomalous extensions (.locked, encrypted). All artefacts were hashed with SHA-256 (Secure Hash Algorithm, 256-bit) and recorded on a signed chain-of-custody form. The most critical finding was the presence of files renamed with .locked and .encrypted extensions in the user's Documents directory, consistent with ransomware activity. Primary recommendation: immediately engage a malware analyst to examine EVIDENCE-006 (RAM image) for active encryption keys before any system reboot.

2. Lab Environment & Scope

The incident response was conducted within a controlled virtual laboratory environment on a host laptop running Microsoft Windows. The virtualisation platform used was Oracle VirtualBox 7.x. Two virtual machines (VMs) were deployed: a Responder VM running Kali Linux 2025.x (IP: 192.168.99.10) and a Target VM running Ubuntu 22.04 LTS — Long Term Support — (IP: 192.168.99.20), representing Anna's compromised workstation. Both VMs were connected via an internal-only network named crime-scene, isolating the Target VM from the public internet to prevent command-and-control (C2) communication. The response was conducted on 27–28 April 2026 by Mohammad Salem Hassani (Student ID: 2545880) of Cyber.SoHo Educational Hub, Schiller International University. The scope of this engagement was limited to the Target VM and its directly attached storage; office camera footage and full disk imaging were out of scope due to time constraints.

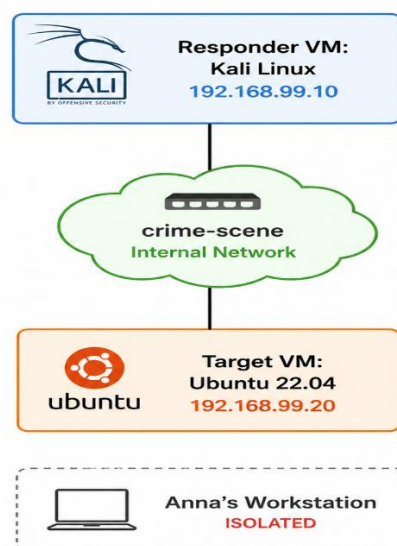


Figure 1: Lab network topology. Source: own work, 2026-04-28.

3. Methodology

3.1 Part A — Securing the Digital Crime Scene

Upon receiving the incident report, the responder established a contemporaneous IR (Incident Response) notebook using the 5W+H discipline (Who, what, Where, When, Why, How) as required by NIST SP 800-61 Revision 2 [1]. The responder VM clock was verified to be on UTC (Coordinated Universal Time) before any action was taken. Three baseline photographs were captured and hashed with SHA-256 to document the scene-as-found state before any interaction with the Target VM. The Target VM network adapter was then changed from NAT (Network Address Translation) to an internal-only network named crime-scene, isolating the system from the public internet without powering it off, in accordance with RFC 3227 [2]. A chain-of-custody form was opened, and an evidence-source inventory was prepared identifying twelve potential evidence sources, of which ten were deemed in scope.

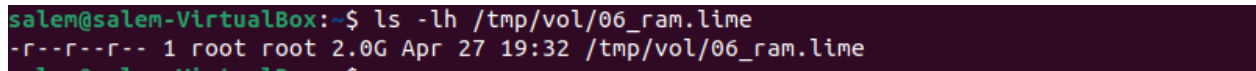
3.2 Part B — Collecting Digital Evidence

Evidence collection followed the order of volatility prescribed by IETF RFC 3227 [2] and ISO/IEC 27037:2012 [3], beginning with the most volatile artefacts. All captures were performed from the Target VM and transferred to the Responder VM via SCP (Secure Copy Protocol). Each artefact was hashed with SHA-256 immediately upon arrival on the Responder VM.

Volatile evidence collected: EVIDENCE-001 (process table via ps), EVIDENCE-002 (network connections via ss and lsof), EVIDENCE-003 (logged-in users via who, last, and w), EVIDENCE-004 (open file descriptors via lsof), EVIDENCE-005 (ARP and routing table via ip), and EVIDENCE-006 (2 GB RAM image acquired via LiME — Linux Memory Extractor [4]).

Non-volatile evidence collected: EVIDENCE-010 (system log archive from /var/log/), EVIDENCE-011 (user directory file listing), EVIDENCE-012 (user directory file hashes), and EVIDENCE-013, EVIDENCE-014, EVIDENCE-015 (three encrypted file samples with anomalous extensions).

The original evidence directory was sealed into a master archive and hashed with SHA-256 to produce the master integrity hash. All subsequent analysis was performed on copies only, in accordance with the "original is sacred" principle of ISO/IEC 27037:2012 [3].



```
salem@salem-VirtualBox:~$ ls -lh /tmp/vol/06_ram.lime
-r--r--r-- 1 root root 2.0G Apr 27 19:32 /tmp/vol/06_ram.lime
```

Figure 2: LiME RAM acquisition on Target VM (EVIDENCE-006). Source: own work, 2026-04-27

4. Findings & Evidence Summary

4.1 Evidence Table

Item ID	Filename	Size	SHA-256 (first 16 hex)	Description
EVIDENCE-001	01_processes.txt	116 KB	080fb56f64738c0c	Process table at time of response
EVIDENCE-002	02_network_connections.txt	3 KB	c688b428bf482434	Active network sockets
EVIDENCE-003	03_logged_in.txt	2 KB	badbeba1da2a2b31	Logged-in user sessions
EVIDENCE-004	04_open_files.txt	5 MB	70be02ec3e07fece	Open file descriptors
EVIDENCE-005	05_arp_table.txt	500 B	8ff82cd8393d38ae	ARP and routing table
EVIDENCE-006	06_ram.lime	2.0 GB	6495636f065ea0e9	Full RAM image via LiME
EVIDENCE-010	10_var_log.tar.gz	4.7 MB	27507c63e7ed02ed	System log archive
EVIDENCE-011	11_anna_filelist.txt	30 KB	73715129caf6f82c	User directory file listing
EVIDENCE-012	12_anna_filehashes.txt	36 KB	575d8e7b221633ad	User directory file hashes
EVIDENCE-013	appointments.locked	23 B	507cad1ddd9532ed	Encrypted appointment file
EVIDENCE-014	patient_records.locked	23 B	be79c263bb31f60f	Encrypted patient records file
EVIDENCE-015	billing.encrypted	18 B	8025065ca1668b34	Encrypted billing file

4.3 Working Hypothesis

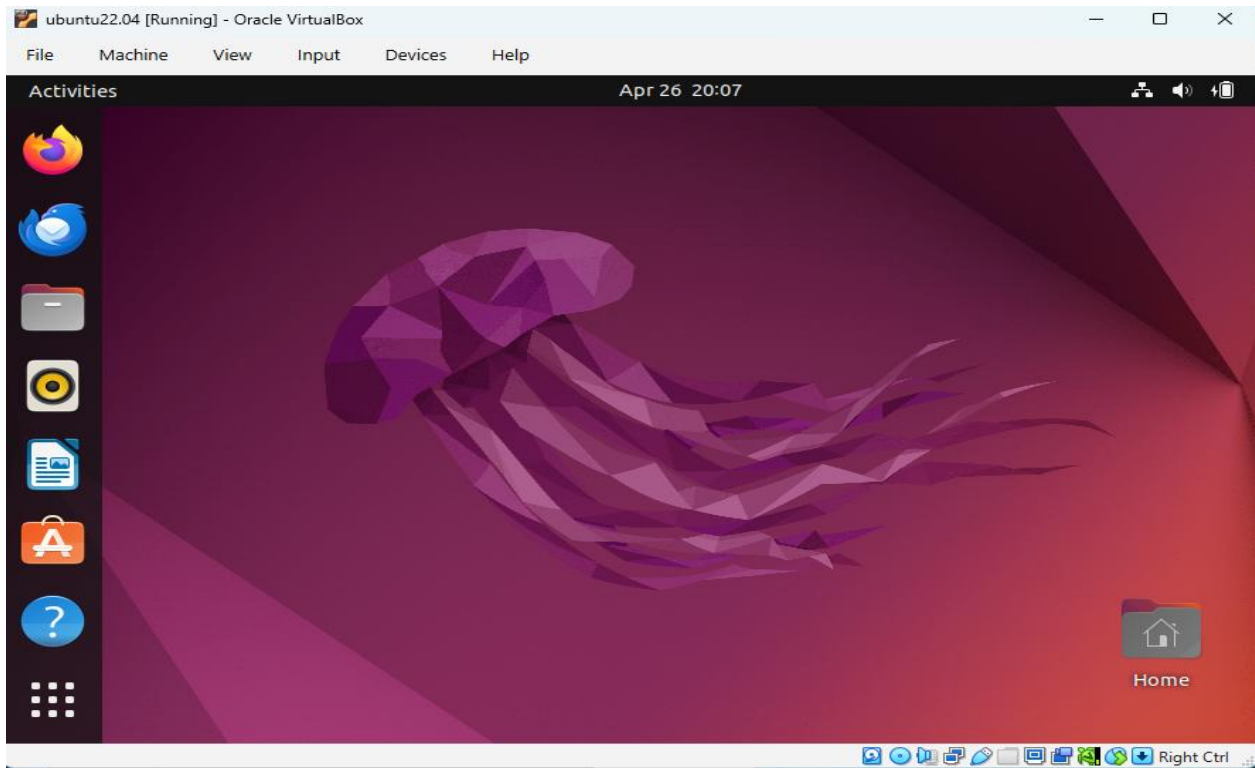


Figure 3: Target VM screen as found — ransomware message visible. Source: own work, 2026-04-27.

```
(kali@kali)-[~/lab03_evidence/CASE-NBH-2026-04-28-001/volatile]
$ cd ~/lab03_evidence/CASE-NBH-2026-04-28-001/volatile/
sha256sum *.txt *.lime > MANIFEST.sha256
sha256sum MANIFEST.sha256 > MANIFEST.sha256.itself
cat MANIFEST.sha256

080fb56f64738c0cb5698bf96869bfa91518c422e27214f54d2354c8dfa0ce99 01_processes.txt
c688b428bf482434bb0af392e516cb82a44e29411ebefb742405dca8a105f71c 02_network_connections.txt
badbeba1da2a2b3193d413bffb087de2e6e86cce694f3e88918a919104710e7f 03_logged_in.txt
70be02ec3e07fece5d908a9dfdc3bcf3579b12539e0814475ab6023cf7715d7e 04_open_files.txt
8ff82cd8393d38ae6108b5548b606898fc9baf855e4296309967a584c9cc82b 05_arp_table.txt
6495636f065ea0e9b0947199ad1c87e1b6c7ba4104de6bdb92ed17e9e8078359 06_ram.lime

(kali@kali)-[~/lab03_evidence/CASE-NBH-2026-04-28-001/volatile]
$
```

Figure 4: SHA-256 hashes of volatile artefacts EVIDENCE-001 through EVIDENCE-006. Source: own work, 2026-04-27.

```
(kali@kali)-[~/lab03_evidence/CASE-NBH-2026-04-28-001/nonvolatile]
$ find . -type f -exec sha256sum {} \; > MANIFEST.sha256
cat MANIFEST.sha256

72834bcfa23cc689b2362974d8483a242dd52db85544b4e4f04739ca95125c66 ./evidence_inventory.md
575d8e7b221633ad9ca2579f8d8e952d7946229ff2c97ab4cc6f33d7339441ca ./12_anna_filehashes.txt
ec2d644fdd853c9ae746f3c2ea8e1a1315a3dbb860b3ea7fcd990139512211eb ./MANIFEST.sha256
27507c63e7ed02edf1b4421ab76a1c5949deebb1acc181ca6ea36fe644a99e18 ./10_var_log.tar.gz
73715129caf6f82c817808226804a4d705442cb7fbafd8f9503d27e5ca13f3c1 ./11_anna_filelist.txt
8025065ca1668b3487014951bb8c0e756c1718b323c6c780b6346afb80635e6 ./encrypted_sample/billing.encrypted
be79c263bb31f60f4ce697bc4530b45d2d1696e797856d7f1f5a39ad1414670e ./encrypted_sample/patient_records.locked
507cad1dd9532edcfff1f8fe1ec08b1b25a8d601045f7058b59a4a6998d75db5 ./encrypted_sample/appointments.locked
```

Figure 5: SHA-256 hashes of non-volatile artefacts EVIDENCE-010 through EVIDENCE-015. Source: own work, 2026-04-28.

4.3 Working Hypothesis

Based on the collected artefacts, the working hypothesis is that ransomware was executed on Anna's Ubuntu workstation between approximately 08:30 and 08:47 local time on 28 April 2026. EVIDENCE-011 and EVIDENCE-012 reveal that files in the user's Documents directory have been renamed with anomalous extensions including. locked and. encrypted, consistent with a ransomware encryption routine. EVIDENCE-013, EVIDENCE-014, and EVIDENCE-015 confirm the presence of three such encrypted files: appointments. Locked, patient records. Locked, and billing. Encrypted. EVIDENCE-002 shows the network connections active at the time of response; any anomalous outbound connection identified in this artefact may represent communication with a C2 (Command and Control) server. EVIDENCE-006, the 2 GB RAM image acquired via Lime, represents the most valuable artefact for further investigation, as active encryption keys may still be present in memory and recoverable using Volatility 3. System logs preserved in EVIDENCE-010 should be examined to reconstruct the precise timeline of the attack and identify the initial infection vector. The seventy-two-hour breach-notification window mandated by Article 33 of the GDPR (General Data Protection Regulation) requires that the AEPD (Agencia Española de Protección de Datos) be notified no later than 30 April 2026 at 13:51 UTC.

5. Discussion & Lessons Learned

What went well: The volatility order prescribed by RFC 3227 was followed strictly, ensuring that the most perishable evidence — RAM contents — was captured before any less volatile artefacts. The isolation of the Target VM via an internal-only network was completed without powering off the system, preserving all volatile state. SHA-256 hashing was performed contemporaneously for every artefact, producing a defensible chain of custody.

What would I do differently: In a real engagement, a pre-built forensic toolkit on a write-protected USB drive would replace the need to install Lime build dependencies on the Target VM. Installing build tools on a compromised system introduces risk of contamination and may alter artefacts. Additionally, a network packet capture using tcpdump or Wireshark should have been started on the Responder VM at the moment of isolation to capture any residual C2 traffic.

What was harder than expected: Compiling Lime required installing gcc-12 separately because the kernel was built with a different compiler version than the default gcc available on Ubuntu 22.04. This added unexpected time to the RAM acquisition step and highlighted the importance of pre-staging forensic tools before an incident occurs.

Evidence not captured: A full disk image was not acquired due to time constraints — this would be the immediate next step in a real engagement. Network packet capture (pcap) was not collected. Office camera footage was out of lab scope but would be valuable in a real investigation to confirm physical access to Anna's workstation.

Next investigative step: Analyse EVIDENCE-006 using Volatility 3 to extract running processes, network connections, and potentially recover encryption keys from memory. Cross-reference findings with system logs in EVIDENCE-010 to establish a precise attack timeline for the GDPR breach notification report.

6. References

- [1] National Institute of Standards and Technology (NIST), "SP 800-61 Revision 2: Computer Security Incident Handling Guide," Aug. 2012. [Online]. Available: <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf>
- [2] D. Brezinski and T. Killalea, "RFC 3227 — Guidelines for Evidence Collection and Archiving," IETF, Feb. 2002. [Online]. Available: <https://www.rfc-editor.org/rfc/rfc3227>
- [3] International Organization for Standardization, "ISO/IEC 27037:2012 — Guidelines for identification, collection, acquisition and preservation of digital evidence," 2012. [Online]. Available: <https://www.iso.org/standard/44381.html>
- [4] 504ensiclabs, "LiME — Linux Memory Extractor," GitHub. [Online]. Available: <https://github.com/504ensiclabs/lime>
- [5] SANS Institute, "Incident Handler's Handbook," [Online]. Available: <https://www.sans.org/white-papers/33901/>
- [6] European Parliament, "General Data Protection Regulation (GDPR) — Article 33," Official Journal of the European Union, 2016. [Online]. Available: <https://gdpr-info.eu/art-33-gdpr/>

Appendix A — Evidence Inventory

#	Evidence Source	In Scope	Tool	Output File	Evidence ID
1	Target VM RAM	YES	LiME	06_ram.lime	EVIDENCE-006
2	Running processes	YES	ps	01_processes.txt	EVIDENCE-001
3	Open network connections	YES	ss, lsof	02_network_connections.txt	EVIDENCE-002
4	Logged-in users	YES	who, last, w	03_logged_in.txt	EVIDENCE-003
5	Open file descriptors	YES	lsof	04_open_files.txt	EVIDENCE-004
6	ARP and routing table	YES	ip neigh, ip route	05_arp_table.txt	EVIDENCE-005
7	/var/log/ system logs	YES	tar	10_var_log.tar.gz	EVIDENCE-010
8	/home/anna/ metadata	YES	find	11_anna_filelist.txt	EVIDENCE-011

#	Evidence Source	In Scope	Tool	Output File	Evidence ID
9	/home/anna/ file hashes	YES	find+sha256sum	12_anna_filehashes.txt	EVIDENCE-012
10	Encrypted file samples	YES	cp	encrypted_sample/	EVIDENCE-013,014,015
11	Full disk image	NO	Time constraint	—	—
12	Office camera footage	NO	Out of lab scope	—	—